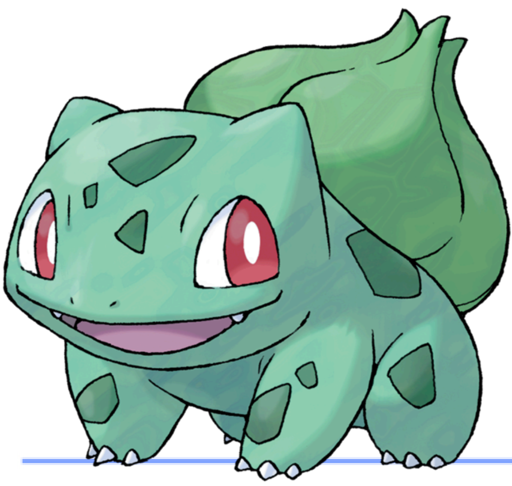
EPITECH

# EPIQUESTS_

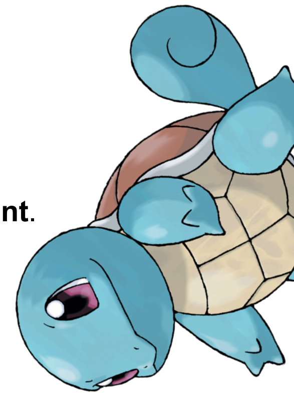
## < UNIX ENVIRONMENTS />

# Summary





# Throughout the entire pool



## EpiGif [PGE / BACH]

Throughout the Piscine, everyone will have the chance to take part in the **EpiGif tournament**.
The goal is simple: **create the best GIF representing Epitech Nice**.

There are only **3 rules to follow**:
- Your GIF must be directly related to Epitech.
- No form of discrimination will be tolerated (this will lead to disqualification).
- Have fun!

Once your GIF is ready, share it on your Discord server with the caption **#EpiGif**.
The winner will be chosen on **Monday, October 27th at 10:00 AM** and will receive **500 points**.

## EpiQuizz [PGE / BACH]

Every **Wednesday afternoon**, during the 5 weeks of the Piscine, you'll be able to team up and join the EpiQuizz. The EpiQuizz is an interactive quiz where you'll play in teams and answer a series of questions. Each week, the theme will change: music, video games, tech culture, and many more!

Here are the rewards:

🥇 1st place: **100 points**

🥈 2nd place: **75 points**

🥉 3rd place: **50 points**

🎖️ Last place: **25 points**


## Team Challenges [PGE / BACH]

On Thursday, **September 25th**, each team chose a challenge.

These challenges allow you to **earn the stickers representing the teams**.
Example: to get the red sticker, you must complete the Red Team's challenge.

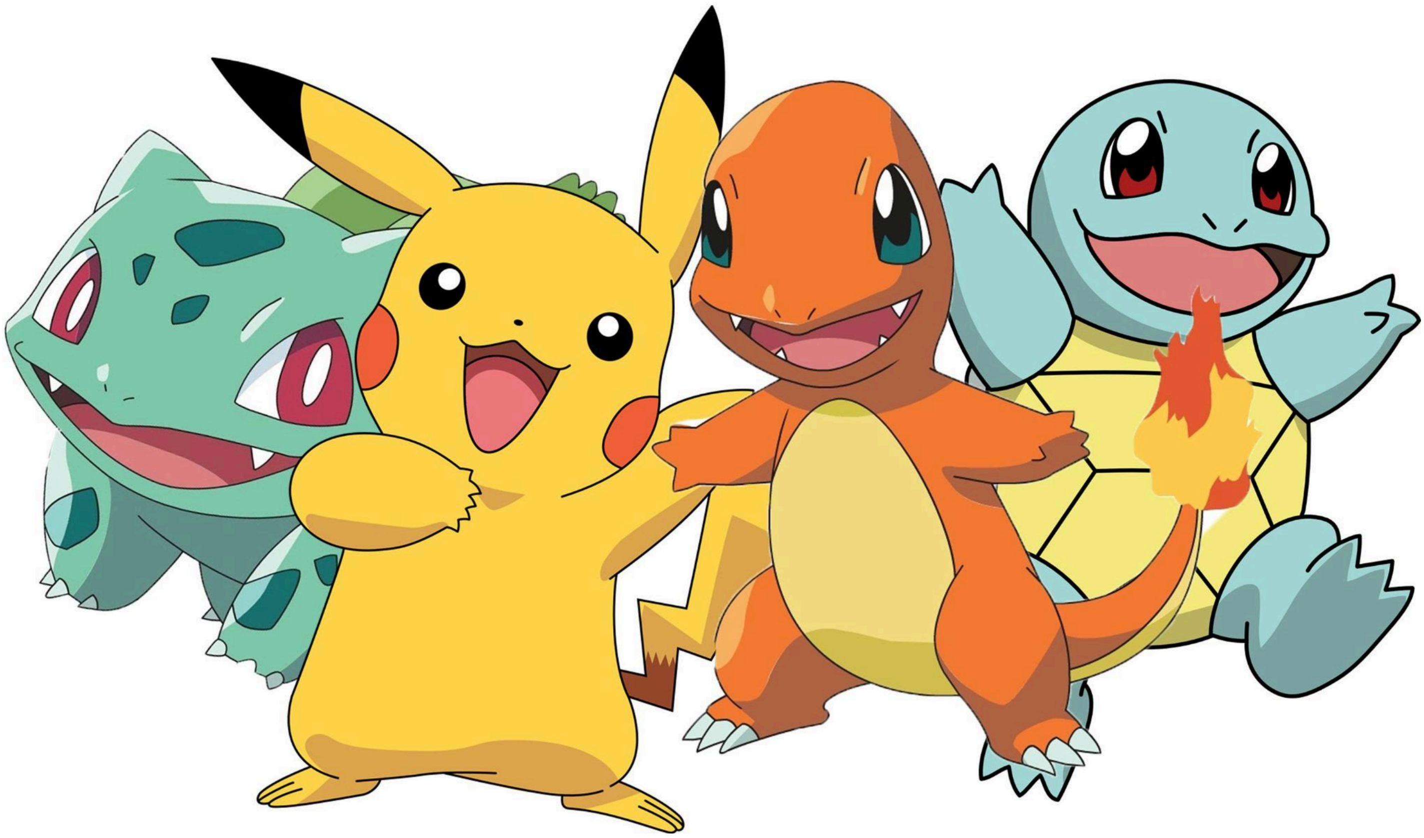
Here are the challenges for each team:

🟢 Green Team → Defend a task in front of the whole class (done by 1 or 2 students) to earn the sticker.

🟡 Yellow Team → Hold a 2-minute plank to get the sticker. After that, the team that lasts the longest earns extra points.

🔴 Red Team → Give a presentation about an AER in groups of 3 (an interview of 5-10 minutes to gather information).

🔵 Blue Team → A coding challenge in 2–5 minutes or code debugging / CodingStyle correction.



## Skill Tokens [Only PGE]

During your C Piscine, you will **validate different skills through the exercises you complete each day**, as checked by Marvin.

Each validated skill will earn you **a colored token**.

There are **8 skills**, and therefore **8 token colors**:
Red - Yellow - Light Green - Dark Green - Light Blue - Dark Blue - Black - White

You can place your tokens into the **white honeycomb** that was given to you at the beginning of the Piscine.

Your goal: **fill every slot in your honeycomb** by validating each skill!

## Daily Coding Challenges [PGE / Soon BACH]

Every day, a coding challenge will take place. **One member from each team will be selected** to participate.

You will have **30 minutes** to complete the given exercise. The challenge will be in **C**, and the solution must both **work correctly** and **follow the Coding Style**.

Each participant who completes the challenge will earn points for their team:

🥇 1st place: **100 points (+ a bonus sticker)**

🥈 2nd place: **75 points**

🥉 3rd place: **50 points**

🎖️ 4th place: **25 points**

⚠️ Be careful: it's possible that **no team will score points** if no one completes the challenge successfully.

## EpiGambling [PGE / BACH]

During the Piscine, you'll have the chance to take part in **fun betting games** (no real money involved, only points).

From wagers to mini-games to silly challenges, you can **propose your ideas to the AERs** for a chance to win… or lose points!

Here are a few example bets:

- 💪 **Arm wrestling against AER Raphaël** → +500 points if you win, but -10 points if you lose.
- ⚔️ **Race between Celenzo (blue team) and Matthis (green team)** → whoever finishes the daily exercises first gives 10 points to the other.
- ♠ **Blackjack between the red team and the yellow team** → the losing team gives 100 points to the winning team.
- 🔎 **Coin hunt** → one team hides a coin in the room, and the first to find it within 5 minutes earns 100 points.

⚠️ All bets **must be approved by an AER** before being played!

# One week - One quest

## Week 1 - Discovering your intranet [Only PGE]

Building projects is great, but it's even better when everyone can understand and use them!
That's where a simple yet powerful markup language comes in: **Markdown**.

Markdown is used everywhere: on Discord, on GitHub, and in many other tools. It allows you to format text easily and in a clean, readable way.

As future developers, you'll use it throughout your entire career-so you might as well start now!

GitHub even supports Markdown files directly. For instance, if you create a **README.md** file in a repository, its content will automatically appear on the project's main page. It's most often used to describe and document the project.

To help you get started with Markdown and to explore Epitech's awesome intranet, your mission is to **find a file that explains how to create a README in the intranet**.

Once you find it, **show it to an AER**, and your team will earn points!

Here are the rewards:

🥇 1st place: **100 points - Lukas SOIGNEUX (Jaune PGE)**

🥈 2nd place: **75 points - Alessandro PARIS (Bleu PGE)**

🥉 3rd place: **50 points - Erwan LO PRESTI (Vert PGE)**

🎖️ Last place: **25 points**

## Week 1.5 - Team card [PGE / BACH]

Your next mission: **create a custom Pokémon card representing your team**.
It must be fully personalized and reflect your group's identity.

Once your card is ready, download it and go see **Andy MALLET** so he can print it for you.

This activity has two goals:
- Help you get to know the administration team at Epitech.
- Strengthen the bonds within your group as you design your Pokémon card together.

Here are the rewards (depending on the speed):

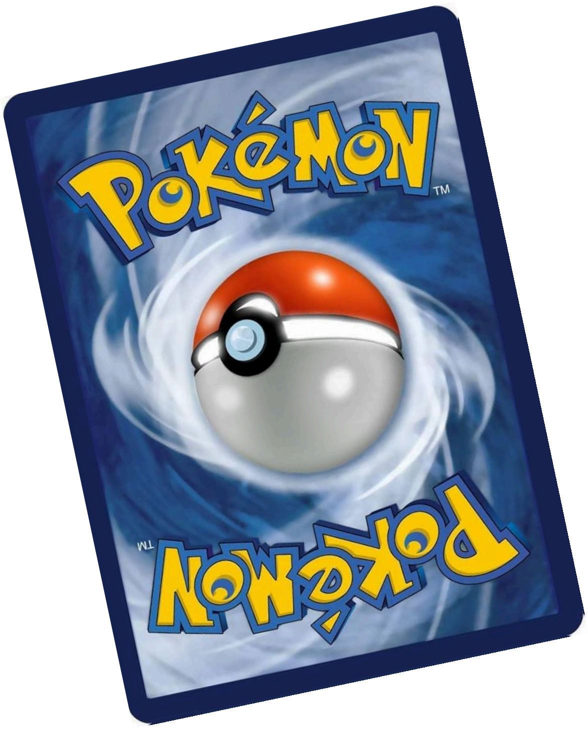
🥇 1st place: **100 points - Equipe Verte**

🥈 2nd place: **75 points - Equipe Rouge**

🥉 3rd place: **50 points**

🎖️ Last place: **25 points**

On top of that, the best card (the one that best represents its team) will be chosen by the teaching staff, and the winning team will receive an **additional 100 points**.



## Week 2 - EpiDrawing [PGE / BACH]

Starting from the second week of the Piscine, the **drawing contest begins**! The idea is simple: **grab a sheet of paper and let your creativity flow**.

⚠️ Important: **no obscene drawings**. Keep it r**elated to the Piscine themes**: Epitech, C, Bash, GitHub, Marvin, or Piscine life.

Once your masterpiece is finished:
1. Don't forget to add your signature or your full name.
2. Get some tape from the Hub.
3. Hang your drawing on a wall in the Ada room.

At the end of the Piscine, all drawings will be voted on, and the **best one will be chosen**. The winner will earn **200 points**!

## Week 3 - EpiSkin [PGE / BACH]

Your next mission: create **a custom Minecraft skin representing your team's Pokémon trainer**. It must be fully personalized and **reflect your group's identity**: colors, style, attitude... make it unique!

But there's a twist: You must create it using the **Mac Minis only**. That's part of the challenge: learn how to use the tools available at Epitech to bring your idea to life!

Once your skin is ready, go show it to **an AER for validation**.

Here are the rewards (depending on the speed):

🥇 1st place: **100 points**

🥈 2nd place: **75 points**

🥉 3rd place: **50 points**

🎖️ Last place: **25 points**

On top of that, the best skin (the one that best represents its team) will be chosen by the teaching staff, and the winning team will receive an **additional 100 points**.

## Week 4 - 3D Model a Staff Member [PGE / BACH]

Your next mission: **model a staff member (an AER or a pedagogy team member) in 3D**.

During week 4 of the Piscine, every team can create a 3D model of a chosen staff member (sculpt, model, texture: **make it recognizable and respectful**). Keep it PG: no obscene or offensive content.

A few rules & tips:
- Work as a team: **planning, modelling, texturing** (divide the tasks).
- The model must be **physically printable** (check manifold, scale, and supports), you'll have to actually print it.
- Once printed, **show the physical print to an AER for validation** (the print proof is mandatory to claim rewards).

Objectives:
- Learn **3D modelling basics** (mesh topology, normals, UVs) and printing constraints.
- **Practice teamwork and presentation skills** (present your process and choices when showing the print).
- Get to **know the staff a little better**, ask them for references/photos if needed (politely).

Here are the rewards:

🥇 1st place (best model): **400 points**

🏅 Every real participant who actually modeled and showed a printed proof to an AER: **200 points each**

The "best model" is **chosen by the teaching staff/AERs based on quality, likeness, printability and creativity**.

Good modelling and may the best (respectful) sculpt win!

## Week 5 - 3D Print Your Team's Pokémon [PGE / BACH]

Your next mission: bring **your team's Pokémon to life in 3D**!

During Week 5 of the Piscine, your goal is to 3D print the Pokémon representing your team using the **3D printers available at Epitech**.

You can:

- **Download an existing 3D model** of your Pokémon from the internet (for example, Pikachu, Bulbasaur, Squirtle, or Charmander).
- Or, if you're feeling **creative, modify or design your own model** before printing!

A few rules & guidelines:

- The model must be printed using **the school's 3D printers only**.
- Ask an AER for assistance if you need help with slicing or printer setup.
- Once printed, **show your 3D Pokémon to an AER for validation**.
- Make sure the model is safe (no sharp edges or broken pieces).
- One print per team, but **everyone should take part in preparing or printing it**!

Objectives:

- Learn **how to use 3D printing tools** available at Epitech.
- Strengthen team collaboration while bringing your mascot to life.
- Leave a physical trace of your team's Pokémon spirit in the hub!

Here are the rewards:

🏆 Every team that successfully prints and shows their Pokémon to an AER: **100 points**

**No ranking this time, every successful print counts!**

So grab your filament, prepare your files, and let your Pokémon take shape!

# One day - One quest [PGE / BACH]

## Day 1 - Display_evolution.sh

Your next challenge is to create a Bash script named **display_evolution.sh**.

This script should take a starter Pokémon (Charmander, Bulbasaur, or Squirtle) as **an argument** and **display its full evolution line**.

Example usage:

```
Terminal
$> ./display_evolution.sh Charmander
Charmander - Charmeleon - Charizard
```

If no argument is provided, or if the given Pokémon is not a valid starter, the script should display:

```
Terminal
$> ./display_evolution.sh Meowth
This is not a starter.
```

Once you make it, **show it to an AER**, and your team will earn points!

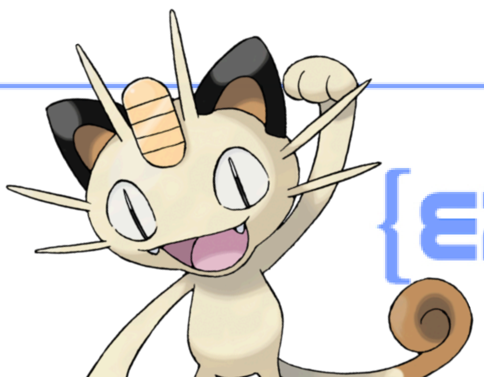
Here are the rewards:

🥇 1st place: **50 points - Lukas SOIGNEUX (Jaune PGE) - Sofia ZGUIRI (Vert BACH)**

🥈 2nd place: **35 points - Alessandro PARIS (Bleu PGE)**

🥉 3rd place: **25 points - Erwan LO PRESTI (Vert PGE)**

🎖️ Last place: **15 points**



## Day 2 - Pokedex.sh

Your next challenge is to create a Bash script named **pokedex.sh**.

This script should **print** out the complete list of the **first 151 Pokémon**. To complete this, you **must use** a **loop** and **a list** (no spamming echo 151 times 😏).

Example usage:

```
Terminal
$> ./pokedex.sh
Bulbasaur
Ivysaur
…
Mew
```

Once you make it, **show it to an AER**, and your team will earn points!

Here are the rewards:

🥇 1st place: **50 points - Lukas SOIGNEUX (Jaune PGE) - Sofia ZGUIRI (Vert BACH)**

🥈 2nd place: **35 points - Erwan LO PRESTI (Vert PGE)**

🥉 3rd place: **25 points**

🎖️ Last place: **15 points**

# Day 3 - Print pokémon

Write a **C function** named:

```c
void my_print_pokemon(int id);
```

This function should **print the team mascot based on the given ID**:
- 0: Pikachu
- 1: Bulbasaur
- 2: Squirtle
- 3: Charmander
- Any other number: unknown

The function must be **implemented using a switch / case**.

Example:

```c
int main(void)
{
    my_print_pokemon(0);
    my_print_pokemon(1);
    my_print_pokemon(2);
    my_print_pokemon(3);
    my_print_pokemon(4);
    return 0;
}
```

Terminal

```
$> ./a.out
Pikachu
Bulbasaur
Squirtle
Charmander
unknown
```

Once you make it, **show it to an AER**, and your team will earn points!

Here are the rewards:

🥇 1st place: **50 points - Lukas SOIGNEUX (Jaune PGE) - Kathleen LACANLALE (Vert BACH)**

🥈 2nd place: **35 points - Alessandro PARIS (Bleu PGE) - Sarra BOUKADIDA (Bleu BACH)**

🥉 3rd place: **25 points - Jessym GADDACHA (Vert PGE)**

🎖️ Last place: **15 points**

## Day 4 - Pokemon level

Write a **C function** named:

```c
void my_pokemon_level(const char *str);
```

This function takes a string as input, where **a Pokémon's name and its level are mixed up**.

Your task is to **reformat it** and **display the Pokémon's name followed by its level.**

Example:

```c
int main(void)
{
    my_pokemon_level("pika25chu");
    my_pokemon_level("carapuce10");
    my_pokemon_level("bulbizarre5");
    my_pokemon_level("mew");
    return 0;
}
```

Terminal

```
$> ./a.out
pikachu 25
carapuce 10
bulbizarre 5
mew
```

Once you make it, **show it to an AER**, and your team will earn points!

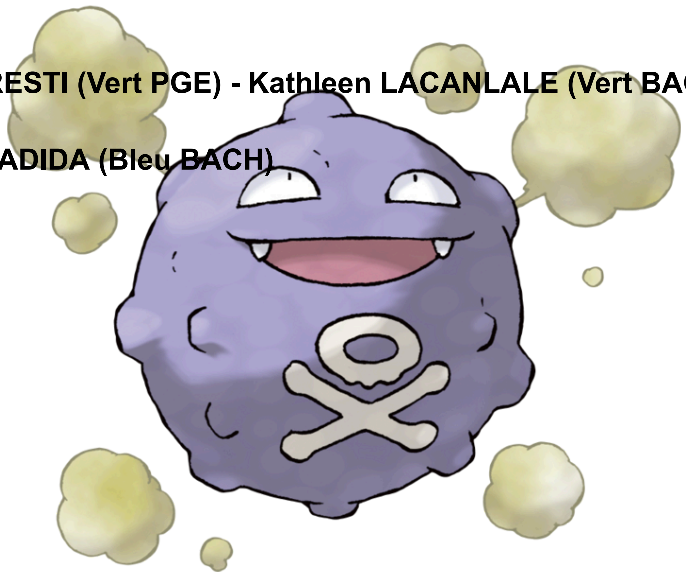
Here are the rewards:

🥇 1st place: **50 points - Erwan LO PRESTI (Vert PGE) - Kathleen LACANLALE (Vert BACH)**

🥈 2nd place: **35 points - Sarra BOUKADIDA (Bleu BACH)**

🥉 3rd place: **25 points**

🎖️ Last place: **15 points**



## Day 5 - Palindrome

Write a **C function** named:

```c
bool my_palindrome_pokemon(char *str);
```

This function should **return true if the given string is a palindrome**, and **false otherwise**.

A palindrome is a word that **reads the same forwards and backwards**.

Example:

```c
int main(void)
{
    printf("eevee -> %s\n", my_palindrome_pokemon("eevee") ? "true" : "false");
    printf("mewtwo -> %s\n", my_palindrome_pokemon("mewtwo") ? "true" : "false");
    printf("girafarig -> %s\n", my_palindrome_pokemon("girafarig") ? "true" : "false");
    printf("pikachu -> %s\n", my_palindrome_pokemon("pikachu") ? "true" : "false");
    return 0;
}
```

Terminal

```
$> ./a.out
true
false
true
false
```

Once you make it, **show it to an AER**, and your team will earn points!

Here are the rewards:

🥇 1st place: **50 points - Alessandro PARIS (Bleu PGE) - Sofia ZGUIRI (Vert BACH)**

🥈 2nd place: **35 points - Erwan LO PRESTI (Vert PGE)**

🥉 3rd place: **25 points**

🎖 Last place: **15 points**

## Day 6 - StrCaseCmp

Write a **C function** named:

```c
int my_strcasecmp(char *str1, char *str2);
```

This function should **compare two strings in a case-insensitive way**, just **like strcmp**, but ignoring uppercase and lowercase differences.

- Return 0 if the strings are equal (ignoring case).
- Return a negative value if str1 < str2.
- Return a positive value if str1 > str2.

Example:

```c
int main(void)
{
    printf("%d\n", my_strcasecmp("Pikachu", "pikachu"));    // 0
    printf("%d\n", my_strcasecmp("Salameche", "SALAMECHE")); // 0
    printf("%d\n", my_strcasecmp("Carapuce", "Bulbizarre")); // != 0
    return 0;
}
```

Terminal

```
$> ./a.out
0
0
1
```

Once you make it, **show it to an AER**, and your team will earn points!

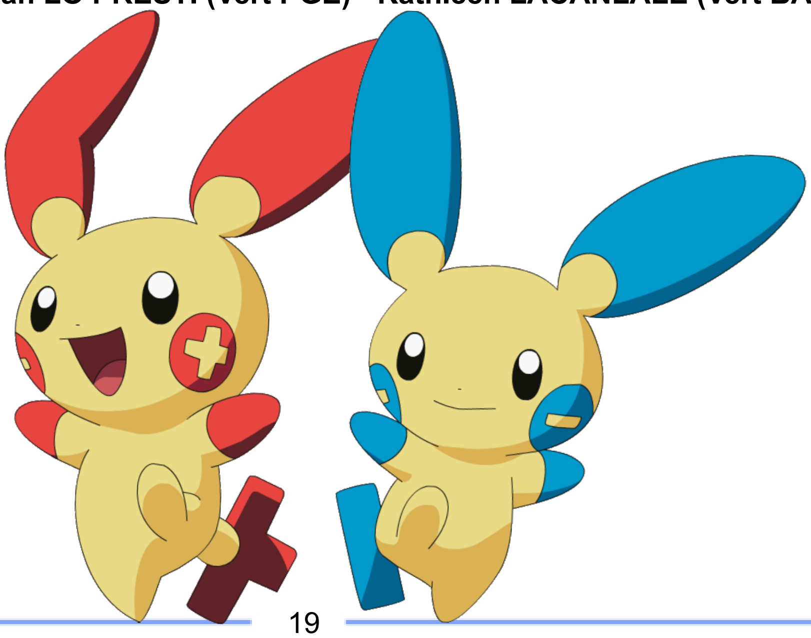
Here are the rewards:

🥇 1st place: **50 points - Erwan LO PRESTI (Vert PGE) - Kathleen LACANLALE (Vert BACH)**

🥈 2nd place: **35 points**

🥉 3rd place: **25 points**

🎖️ Last place: **15 points**



## Day 7 - Print pokémon 2

Write a **C program** named **my_print_pokemon** and used like that :

```
Terminal
$> ./my_print_pokemon <size> <name> <speed>
```

The program takes 3 arguments **in any order**:
- A Pokémon name (char *)
- A size (M, L, XL)
- A speed (int)

Your program must **print the sentence**:
*{name} is {size} and has a speed of {speed}*

⚠️ The difficulty is that the parameters can be passed in **any order**, but your program must still print the correct sentence.

Example:

```
Terminal
$> ./my_print_pokemon Pikachu M 90
Pikachu is M and has a speed of 90

$> ./my_print_pokemon XL Bulbasaur 45
Bulbasaur is XL and has a speed of 45

$> ./my_print_pokemon 65 Charmander L
Charmander is L and has a speed of 65
```

Once you make it, **show it to an AER**, and your team will earn points!

Here are the rewards:

🥇 1st place: **50 points**

🥈 2nd place: **35 points**

🥉 3rd place: **25 points**

🎖️ Last place: **15 points**

## Day 8 - Pokémon to pokedex

Write a **C program** named **my_pokemons_to_pokedex** and used like that :

```
Terminal
$> ./my_pokemons_to_pokedex <pokemon1> <pokemon2> <pokemon3> ...
```

The program should:

- Take a list of Pokémon names as arguments.
- **Store them in an array**.
- **Convert all names to uppercase**.
- Print the full array at the end.

Example:

```
Terminal
$> ./my_pokemons_to_pokedex Pikachu Bulbasaur Charmander Squirtle
→ PIKACHU BULBASAUR CHARMANDER SQUIRTLE
```

Once you make it, **show it to an AER**, and your team will earn points!

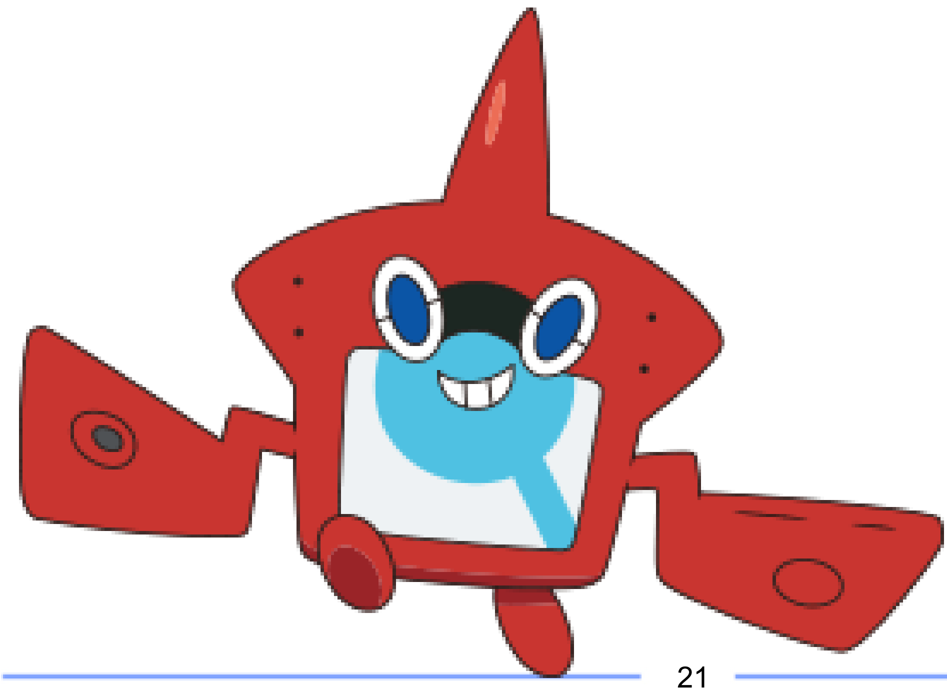
Here are the rewards:

🥇 1st place: **50 points - Erwan LO PRESTI (Vert PGE)**

🥈 2nd place: **35 points - Alessandro PARIS (Bleu PGE)**

🥉 3rd place: **25 points**

🎖️ Last place: **15 points**



# Day 9 - Pokemon structure

Create a **C structure** named **pokemon_t**.

The structure must contain:
- **char *name** → the Pokémon's name
- **char *type** → the Pokémon's type (e.g., Fire, Water, Grass…)
- **char *size** → the Pokémon's size (M, L, XL)
- **int speed** → the Pokémon's speed

Then, write a constructor function:

```c
pokemon_t *create_pokemon(char *name, char *type, char *size, int speed);
```

Example:

```c
int main(void)
{
    pokemon_t *pikachu = create_pokemon("Pikachu", "Electric", "M", 90);
    printf("%s is an %s Pokémon, size %s, speed %d\n",
        pikachu->name, pikachu->type, pikachu->size, pikachu->speed);
    free(pikachu);
    return 0;
}
```

Terminal

```
$> ./a.out
Pikachu is an Electric Pokémon, size M, speed 90
```

Once you make it, **show it to an AER**, and your team will earn points!

Here are the rewards:

🥇 1st place: **50 points - Erwan LO PRESTI (Vert PGE) - Sarra BOUKADIDA (Bleu BACH)**

🥈 2nd place: **35 points - Alessandro PARIS (Bleu PGE) - Kathleen LACANLALE (Vert BACH)**

🥉 3rd place: **25 points**

🎖️ Last place: **15 points**

## Day 10 - Sort pokémon

Write a **C function** named:

```c
void my_sort_pokemons(pokemon_t **pokemon_list, int size);
```

This function should sort **an array of pokemon_t structures** in **alphabetical order by name**.

Example:

```c
int main(void)
{
    pokemon_t pikachu = {"Pikachu", "Electric", "M", 90};
    pokemon_t bulbasaur = {"Bulbasaur", "Grass", "L", 45};
    pokemon_t charmander = {"Charmander", "Fire", "M", 65};
    pokemon_t squirtle = {"Squirtle", "Water", "M", 43};
    pokemon_t *list[4] = {&pikachu, &bulbasaur, &charmander, &squirtle};

    my_sort_pokemons(list, 4);
    for (int i = 0; i < 4; i++) {
        printf("%s\n", list[i]->name);
    }
    return 0;
}
```

Terminal

```
$> ./a.out
Bulbasaur
Charmander
Pikachu
Squirtle
```

Once you make it, **show it to an AER**, and your team will earn points!

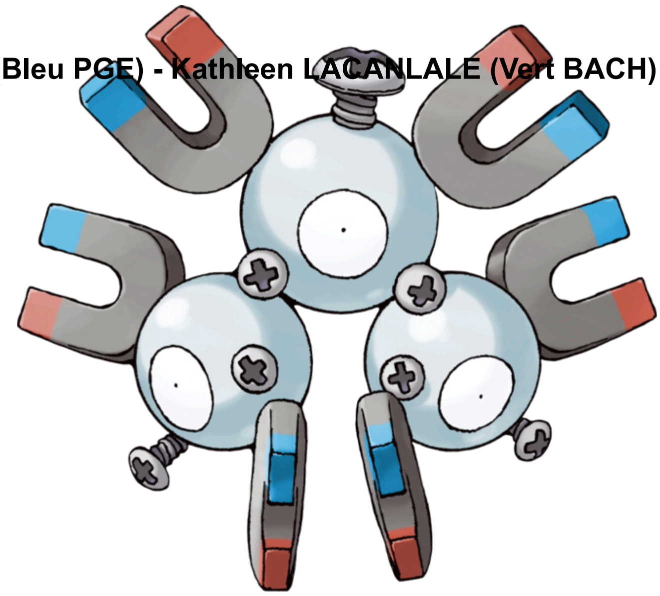
Here are the rewards:

🥇 1st place: **50 points - Alessandro PARIS (Bleu PGE) - Kathleen LACANLALE (Vert BACH)**

🥈 2nd place: **35 points**

🥉 3rd place: **25 points**

🎖️ Last place: **15 points**



# Day 11 - Apply to pokedex

Write a **C function** named:

```
void my_apply_pokedex(linked_list_t *list, int multiplicator);
```

This function should go through a **linked list of Pokémon** (pokemon_t) and **multiply the speed** of each Pokémon by **the given multiplicator**.

Example:

```
int main(void)
{
    pokemon_t pikachu = {"Pikachu", "Electric", "M", 90};
    pokemon_t bulbasaur = {"Bulbasaur", "Grass", "L", 45};
    pokemon_t charmander = {"Charmander", "Fire", "M", 65};
    linked_list_t c = {&charmander, NULL};
    linked_list_t b = {&bulbasaur, &c};
    linked_list_t a = {&pikachu, &b};

    my_apply_pokedex(&a, 2);
    for (linked_list_t *tmp = &a; tmp != NULL; tmp = tmp->next) {
        printf("%s -> speed %d\n", tmp->pokemon->name, tmp->pokemon->speed);
    }
    return 0;
}
```

Terminal

```
$> ./a.out
Pikachu -> speed 180
Bulbasaur -> speed 90
Charmander -> speed 130
```

Once you make it, **show it to an AER**, and your team will earn points!

Here are the rewards:

🥇 1st place: **50 points - Kathleen LACANLALE (Vert BACH)**

🥈 2nd place: **35 points**

🥉 3rd place: **25 points**

🎖️ Last place: **15 points**

# Day 12 - Save to pokedex

Write a **C program** named: my_save_to_pokedex.

This program should take as arguments **a list of Pokémon names** and **save them into a file** named **pokedex.txt**. If the file already exists, it should **append** the new Pokémon names to the file (not overwrite it). Each Pokémon name must be written **on a new line**.

Example:

```
Terminal
$> ./my_save_to_pokedex pikachu bulbizarre salamèche carapuce
$> cat pokedex.txt
pikachu
bulbizarre
salamèche
carapuce
$> ./my_save_to_pokedex dracaufeu ronflex
$> cat pokedex.txt
pikachu
bulbizarre
salamèche
carapuce
dracaufeu
ronflex
```

Once you make it, **show it to an AER**, and your team will earn points!

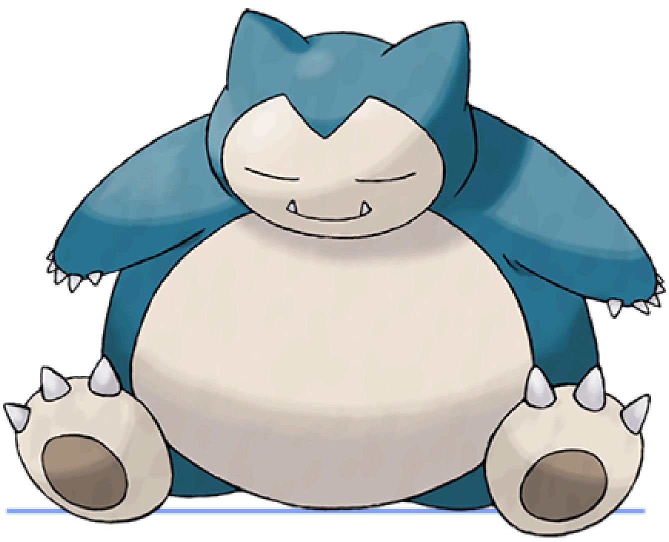
Here are the rewards:

🥇 1st place: **50 points - Kathleen LACANLALE (Vert BACH)**

🥈 2nd place: **35 points**

🥉 3rd place: **25 points**

🎖️ Last place: **15 points**



# Day 13 - Draw pokémon

Write a **C program** named: my_draw_pokemon.

The program must **open a window** and **draw a Pokémon sprite** (from an image file) using **CSFML (or with Pillow for BACH)**.

The program should load the image as a texture, create a sprite, display it centered in the window and keep the window open until the user closes it (or presses Escape).

If no argument is provided, print a usage message and exit.

Example:

```
Terminal
$> ./my_draw_pokemon <pokemon_image.png>
```

Once you make it, **show it to an AER**, and your team will earn points!

Here are the rewards:

🥇 1st place: **50 points - Kathleen LACANLALE (Vert BACH)**

🥈 2nd place: **35 points**

🥉 3rd place: **25 points**

🎖️ Last place: **15 points**